



JAIPUR NATIONAL UNIVERSITY, JAIPUR

BTEBE205 / ECS204

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Roll No.: 0082021020026

BTEBE205 / ECS204

School of Engineering & Technology

B.Tech. II Sem : Biomedical / Biomedical +MBA /

Civil / Elecrical / Electronics / Mechanical / Biotechnology /

Biotechnology + MBA / Chemical / Computer Science /

Food Technology / CSE (AIML) / CSE (Cyber Security)

II SEMESTER END EXAMINATION: MAY 2022

ENGINEERING MECHANICS

Time: 3 Hrs.

Max. Marks: 70

INSTRUCTIONS

Attempt all questions. All questions carry equal marks.

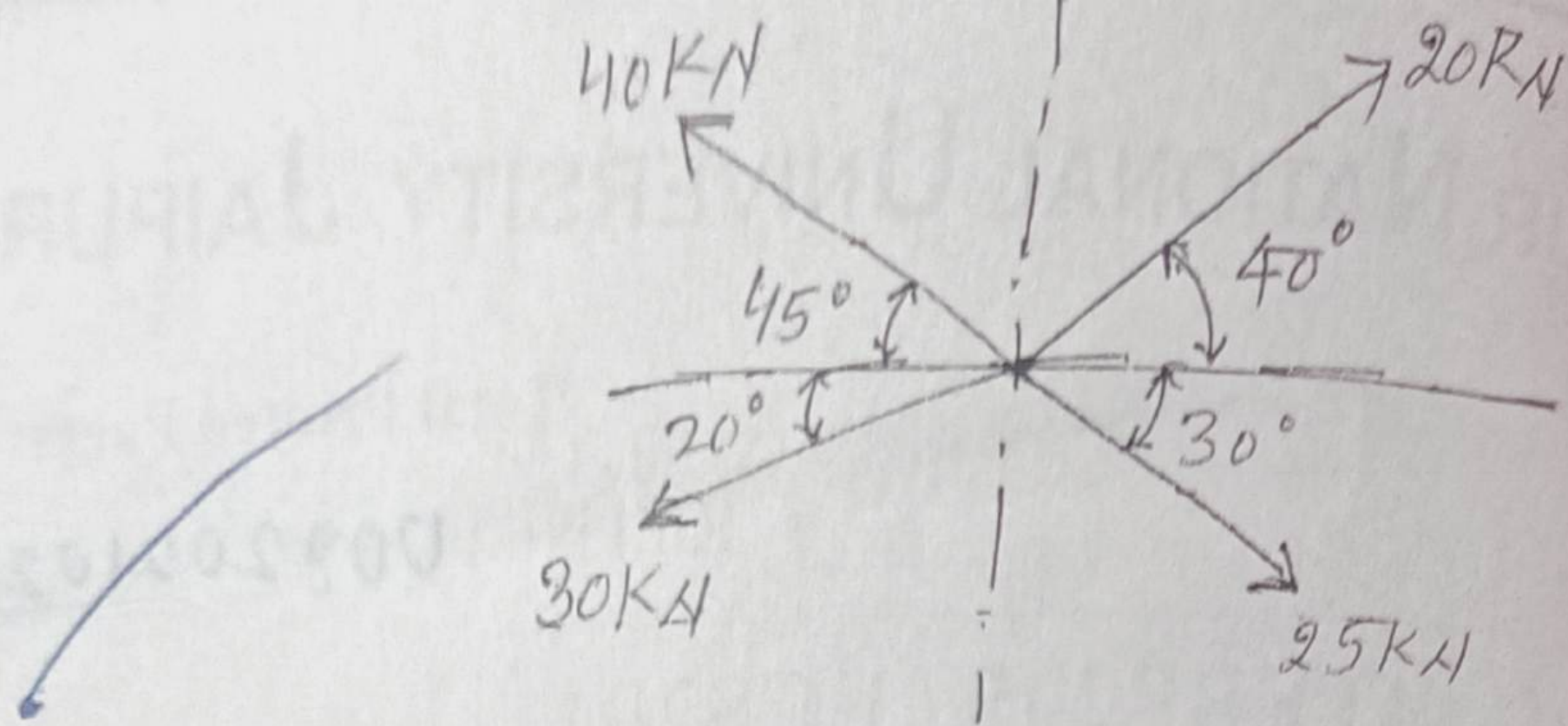
UNIT-I

Q.1 (a) Write short notes on:

(i) Parallelogram Law of Forces 3

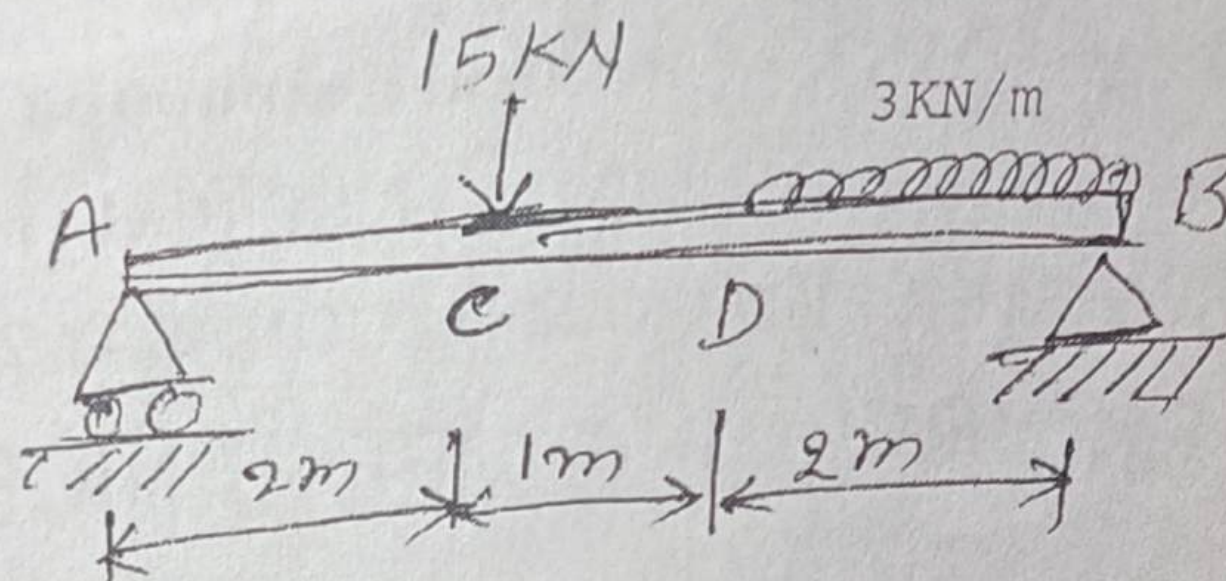
(ii) Lami's Theorem 4

(b) Find the magnitude and direction of the resultant of the given system of forces. 7

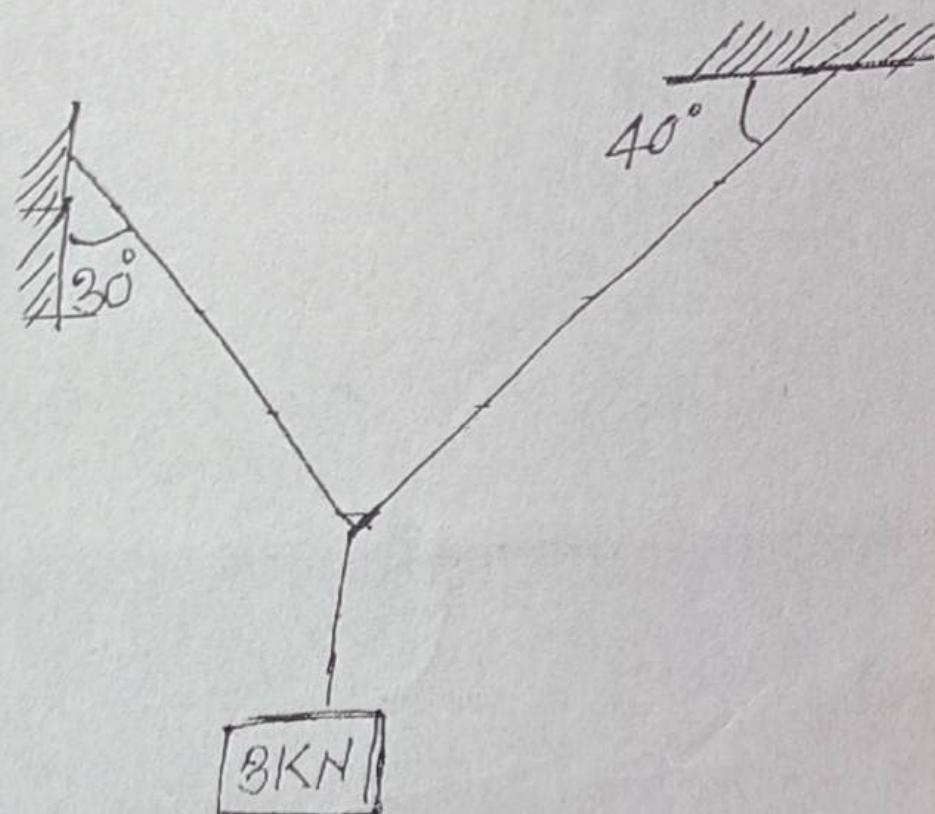


OR

Q.1 (a) Determine the Reaction at Point A and B.



(b) Find the tension in the two ropes in the given system of forces.



UNIT-II

Q.2 (a) Define the principle of virtual work and prove $\delta W = 0$.

A ball is thrown vertically upward with a velocity of 29.4 m/sec from the top of the tower 34.3 m high. Find the total time taken by the ball to reach the floor of the tower.

7

UNIT-V

(a) Write short notes on:

(i) Work

4

(ii) Power

3

(b) Determine the power of an engine required to lift 3×10^3

7

kg water per minute through a height of 20 m. If the

water is coming out with a velocity 5 m/sec then what is

the power?

OR

Q.5 (a) Explain energy.

7

(b) Derive a relation between work and energy.

7

- (b) In an open belt drive the tension on the tight side is 2000 N. The angle of lap is 135° and coefficient of friction is 0.25. Find the tension on the slack side.

OR

- Q.3 (a) A ladder of 10 m length and 4KN weight is placed against a smooth vertical wall with 40° angle from horizontal. In this position, the ladder is just about to slip. Determine the coefficient of friction between the ladder and the floor.

- (b) Derive the tension ratio in flat belt is $\frac{T_1}{T_2} = e^{\mu\theta}$.

UNIT-IV

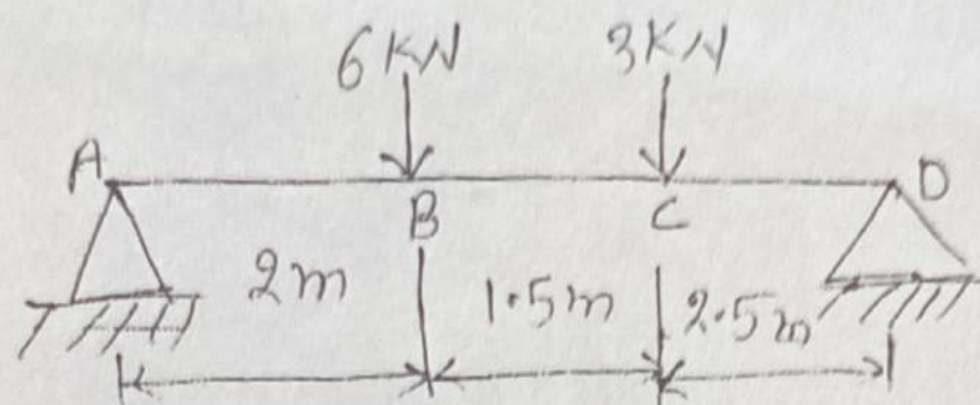
- Q.4 (a) Derive First, Second and Third equations of motion.
- (b) A ball is projected from ground at an angle of 60° with the horizontal, with a velocity of 30 m/sec. Find the distance of the ball from the point of projection, after 2 sec. and its height above the ground.

OR

- Q.4 (a) Derive an expression for the range, maximum height and time of flight for a projectile motion.

- 1) Find the reaction forces using the Principle of virtual work of the given system.

7

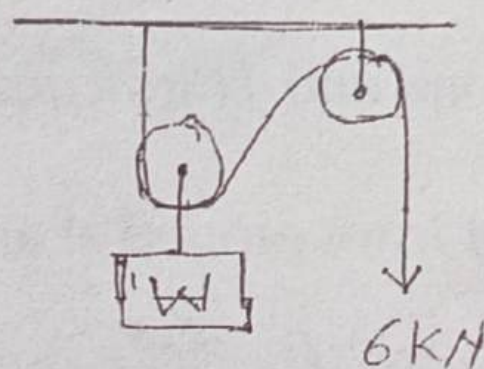


OR

- 2 (a) A uniform ladder of 6m length and 2kN weight rest on a rough floor and smooth wall making an angle 30° with the horizontal. Find the Friction force on the floor using the Principle of virtual work.
- (b) Find the value of W by the principle of virtual work. If the system is in equilibrium.

7

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UNIT-III

Q.3 (a) Write short notes on:

- (i) Angle of Friction
- (ii) Angle of Repose

3

4